IntelliTrack 2.0 - Full Conversation Transcript & Architecture Direction

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User Context: Capstone Project Overview

User explained that the project is a Capstone Deliverable Submission Tracker System using Spring Boot Java 21, React JSX, MySQL/Supabase, and Gemini API. Current roles: - Student - Adviser - Coordinator - Admin Initial concerns included: - broken adviser review workflow - unclear analytics logic - disconnected system workflows - overly broad architecture

Assistant Direction: Refocus the System

The assistant identified that the system was becoming too broad and recommended focusing the project identity into: 'An Intelligent Submission and Compliance Tracking System' Instead of: - LMS platform - grading platform - revision workflow system - approval platform Core focus became: - submission tracking - compliance monitoring - adviser monitoring - coordinator oversight

Coordinator Analytics Requirements

User explained that the coordinator specifically wants operational oversight: Examples: - Group A: 3 submitted on time 1 submitted late 1 missing The assistant explained this is operational compliance analytics, not generic business analytics.

Suggested Analytics Direction

Assistant proposed: - compliance score - late tracking - missing tracking - adviser workload - group risk assessment Example: Group A Compliance Score = 65 Risk Level = HIGH

AI Architecture Recommendation

Assistant emphasized: Backend computes intelligence. AI only explains it. Backend responsibilities: - risk scoring - compliance metrics - late detection - missing tracking AI responsibilities: - reminders - summaries - analytics explanations

System Workflow Evolution

Original workflow: SUBMITTED UNDER_REVIEW APPROVED REJECTED NEEDS_REVISION Final simplified workflow: PENDING SUBMITTED_ON_TIME SUBMITTED_LATE NOT_SUBMITTED COMMENTED Adviser role simplified into monitoring and commenting only.

Architectural Crisis & Refactor

User reported that changing statuses broke multiple system modules. Assistant diagnosed: - excessive coupling - DTO inconsistencies - dead workflow remnants - frontend/backend mismatch Recommendation: perform a controlled vertical refactor.

Refactor Strategy Proposed

Assistant recommended rebuilding vertically by workflow: 1. Student workflow 2. Adviser workflow 3. Coordinator workflow 4. Admin cleanup 5. Final alignment Instead of random isolated fixes.

Student Workflow Requirements

User clarified desired student features: Student dashboard must contain: - AI smart reminders - total deliverables - submission progress - submission history - deadlines - compliance risk indicators

Adviser Workflow Requirements

Adviser pages should: - show assigned groups - display submission progress - display late/missing submissions - allow comments Adviser should NOT: - approve submissions - reject submissions - manage revisions

Coordinator Workflow Requirements

Coordinator dashboard should include: - total students - total groups - total advisers - total submissions - AI institutional reminders - analytics dashboards Additional coordinator requirements: - classlist import - auto-group generation - automatic student assignment - submission tracker - analytics engine

Auto Grouping Logic

User specifically requested: When importing classlists: - groups should auto-create - students should auto-assign - classlist views should display group assignments

Milestone Logic

User requested section-specific deadlines. Example: IT411 Section A: Chapter 1 deadline = May 25 IT411 Section B: Chapter 1 deadline = June 1

Security & Stability Hardening

User shared detailed implementation progress involving: - JWT secret validation - stateless session policy - IDOR protection - transactional upload handling - cache eviction - AI timeout protection - query optimization - logging standardization

AI Agent Development Context

User requested permanent markdown documentation for AI agents. Assistant proposed: - AI_SYSTEM_CONTEXT.md - AI_DEVELOPMENT_RULES.md - IMPLEMENTATION_PHASES.md - VALIDATION_CHECKLIST.md - ANALYTICS_LOGIC.md These files become permanent context for: - Claude Sonnet - Copilot - Trae AI

Final Architectural Direction

Final agreed system identity: IntelliTrack 2.0 An Intelligent Submission and Compliance Tracking System for Academic Capstone Projects Final architectural principles: - submission-first architecture - compliance-based analytics - monitoring-only adviser workflow - coordinator institutional oversight - AI-assisted intelligence - backend-computed analytics